

Privacy and Security Audit of Mobile Applications: A Comparative Study of Android and iOS Using Automated Analysis

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Abstract— Mobile applications have started to process the user's personal sensitive information, making it a critical concern. Even after the dominance of Android and iOS, quantitative comparisons of the real world privacy risks are limited. This research aims to analyse popular mobile applications across both the platforms by using Mobile Security Framework (MobSF) and examine permissions, tracking libraries, WebView behavior, cryptographic practices, network security configuration and platform specific privacy mechanisms. In order to get a fair comparison, we derive standardized metrics and propose the Privacy Risk Index (PRI) using Principal Component Analysis (PCA). This analysis also figures out privacy risks in pre-installed and user-installed applications in Android. The final result shows that in iOS the platform-specific privacy measures showed higher aggregate PRI scores while the Android applications showed greater variability by tracking permissions, doing risk assessment for users and making policies.

Keywords—android, iOS, privacy, security, static analysis, privacy risk index.

I. INTRODUCTION

Smartphone applications are used in our everyday life as they support our everyday tasks like communication, commerce, healthcare, navigation and entertainment. Since everything has its own pros and cons, these applications have started to access users' personal information like location, contacts, biometrics and behaviour patterns which is substantially raising concerns regarding the user's privacy [1], [2]. Malicious practices like data breaches, unauthorised tracking of devices and excessive permissions have emphasized the urgent need for a scalable evaluation of the mobile application system [3].

Android focuses more on extensibility by allowing the developers for a greater flexible environment by introducing the heterogeneous security practices [4] whereas iOS is a more controlled ecosystem which has new policies, mechanism and privacy control such as App Tracking Transparency (ATT) [5]. These differences allow for architectural differences between the two.

Various aspects of mobile security were examined in prior research that included permission misuse [6], insecure network communication, cryptography weaknesses and third party integration [7]. These studies help in providing some valuable insights to limit the privacy concerns. Manually risk scoring methods are used but they may introduce some subjective bias.

In order to address the limitations, we reviewed static analysis of mobile applications on both the platforms, using Mobile Security Framework (MobSF) [8]. We try to find out some relevant security features like the permissions, trackers and analytics SDK presence, JavaScript behaviour, cryptographic practices and some platform related information such as ATT on iOS.

We formulate a Privacy Risk Index using Principal Component Analysis (PCA) [9] and directly address the risk dimensions from the observed data by enabling statistical aggregation of privacy and security factors. This approach allows a robust comparison between Android and iOS applications as well as between the user-installed and pre-installed Android applications.

Apart from the static technical analysis, we incorporate the user perception by relating some privacy risks with the sentiment based trust scores that are derived from Google Play Store reviews. It was often seen from the previous studies that user awareness was misaligned with the actual security properties [10]. Hence by combining static analysis with the user sentiment, a more comprehensive understanding of technical privacy risks can be done [26].

II. BACKGROUND WORK

Android is the primary focus of most previous research due to its open nature and iOS is often ignored due to its closed system. This open nature also brings unusual security risks with it, especially in budget smartphones where pre-installed applications have access and permissions to perform critical system level tasks. This is often missed due to limited examination of system / pre-installed apps [11]. On the other

hand less is known about the on device applications on iOS due to challenges in accessing the app files.

Many studies have explored differences in user security behaviours between Android and iOS users and found that platform differences also affect how people respond. iOS users are sensitive to the cost of security actions due to their closed ecosystem, while Android users navigate more side loading risks [12].

This issue extends to specialized domain specific apps too, like mobile healthcare applications bring significant security risks due to their data practices [13]. Many automated methods are being developed to detect privacy and security issues in Android applications like VioDroid-Finder, which found that apps often lie in their privacy policies [15].

Table I. presents a comparative analysis of current studies to highlight limitations of current research.

TABLE I. COMPARISON OF CURRENT RESEARCH

S.No.	Author name [Year of publication]	Focus	Methodology	Findings	Limitations
1.	[15] Chen et al. (2024)	Evaluation of compliance and consistency of Android apps.	Automated method called VioDroid-Finder	Found many violations and inconsistencies reflecting the current widespread privacy violation issues.	Only policy and code comparison and no comparative audit of android and iOS.
2.	[14] Kishani & Das (2025)	Android(U.S. and Non-U.S. E-commerce apps)	MobSF, AndroBugs, and RiskInDroid	Found that 92% apps used unsecured HTTP connections and average MobSF security score of 40.92/100	Single domain focus (E-commerce apps) only and lacks comparative analysis with iOS.
3.	[13] Stevenson & Das (2025)	Android mHealth apps	MobSF, RiskInDroid, and OWASP Mobile Audit	Found that 26.1% apps request fine-grained location without disclosure, 18.3% initiate calls silently, and 73 send SMS without notice, 49.3% still use deprecated SHA-1 encryption and 42 transmit unencrypted data.	Limited to medical apps and no comparison with the iOS ecosystem.
4.	[16] Hu et al. (2023)	Android & iOS	Study why current developers use different on-device models for one app between iOS and Android	Successfully exploit the vulnerabilities of on-device models to attack real-world iOS apps	Focuses only on on-device models used by the apps.
5.	[17] Khatun et al. (2025)	Android	AndroByte, an AI driven tool that applies bytecode summarization and LLM reasoning to generate dynamic dataflow call graphs.	AndroByte can produce dynamic dataflow call graphs with a high F-score (~89%)	Focuses on privacy leaks(contacts, mic etc.) in android apps and does not compare with iOS apps.
6.	[11] Diallo et al. (2025)	Pre-installed Android applications on low-cost African devices.	PiPLAnD (a static analysis pipeline)	Found that 9% apps disclose sensitive data, 16% expose critical components without sufficient safeguards, and many present additional risks: 226 apps execute privileged or dangerous commands and 79 apps interact with SMS messages (read, send, or delete).	Limited to pre-installed apps on seven low-cost devices sold in Africa.
7.	[18] Garg et al. (2021)	Android & iOS	Analyzes data for the period 2015-2019 and leverages the security triad i.e. CIA (Confidentiality, Integrity, Availability) to compare both the operating systems.	Found that Android is more susceptible to security breaches and malware attacks as compared to iOS.	Uses historical CVE data and lacks application level behavior and permission usage analysis.
8.	[19] Ciaramella et al. (2025)	Android	Federated Machine Learning based approach.	Found that federated models achieve 0.873-0.877 malware detection accuracy and centralized training achieves higher accuracy of ~0.944.	Focuses on image based representations of APKs and malware classification and does no risk assessment at application level.
9.	[20] Khedkar & Bodden (2024)	Android	Identify personal data sources in android app code using static analysis framework and reasons	Highlights use of static analysis to explain personal data flows in android apps and identify challenges to track data manipulation beyond simple taint flows.	Does not compare the findings with the iOS ecosystem.

			about how that data is manipulated.	Pseudonymization functions like MessageDigest and Cipher were detected.	
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While extensive audits exist for Android, like the automated tool - VioDroid Finder, or AndroByte, which applies bytecode summarization to generate dynamic dataflow call graphs with a 89% F-score, there is a lack of comparative studies that apply the same metrics to iOS applications.

III. METHODOLOGY

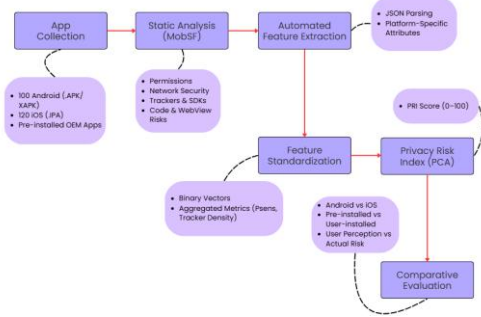


Fig. 1. Block Diagram of Process Flow

A. APK and IPA Collection

We collected 100 Android applications and 120 iOS applications, of which 42 were common across both platforms. The common applications enable direct cross-platform comparison between Android and iOS, while the remaining application supported platform-specific behaviour analysis. The number of common applications remained limited due to restrictions of both .APK and .IPA files, primarily due to distribution and licensing constraints imposed by application developers and the platform itself.

The Android application packages were collected from ApkMirror [21] and iOS applications were extracted as decrypted .IPA files using the iOS management tool iMazing [22], which enabled access to stable application versions for analysis. Selection criteria for these applications were based on their popularity, accessibility and various category diversity, ensuring coverage across multiple functional domains. Some OEM Android applications that come pre-installed, were also included to broaden the scope of our analysis.

B. Experimental Setup

Static analysis was performed using MobSF version 4.4.4, which is an automated analysis platform. It was deployed locally using Docker to ensure environmental consistency and reproducibility. Custom Python automation scripts were developed to streamline analysis. These scripts automated reports retrieval and structured storage of the analysis output.

C. Analysis Procedure

The study relied exclusively on static analysis without executing the applications. The analysis procedure followed the outlined steps below:

- For Android applications, .APK files were not always available. In such cases, .XAPK or .APKM bundles [23] were obtained and converted to ZIP archives and the corresponding base.apk file was extracted manually.

- The extracted .APK Android files and .IPA iOS files were then uploaded to MobSF for analysis.
- Upon completion of the analysis, the automated scripts were executed to retrieve the JSON and MobSF security scorecard for each application.
- Separate parsing scripts were applied to Android and iOS reports to extract relevant security and privacy features.
- The extracted features were then stored in excel sheets for further analysis.

Table II. lists the features that were extracted from the JSON files.

TABLE II. DATA FEATURE EXTRACTION

Feature	Android Attributes
Application Metadata	App Name, Package Name / Bundle Identifier, Version, Hash, Target SDK Version
Permissions	Total permissions, dangerous permissions, Specifics: Camera, Mic, Location, Contacts, SMS
Network Security	Network Security Config Present(Android only), Cleartext Traffic Allowed(Android only), Certificate Pinning(Android only), ATS Exception Present(iOS only)
Tracking & Privacy	Tracker Count, Tracker Names, Ad SDK Count, Analytics SDK Count, PII Collected, App Tracking Transparency(ATT) required(iOS only), Uses IDFA(iOS only)
Code & Components	WebView JavaScript Enabled, Dynamic Code Loading(Android only), Hardcoded Secrets(Android only), Exported Providers Count(Android only), Custom URL Schemes(iOS only)

D. Evaluation Metrics

In order to guarantee the integrity of various platforms, binary vectors were standardized into boolean format. A few security concepts were mapped to some platform-particular descriptions like the Location access was defined by combining various fine-grained permissions. These permissions are triggered when an Android or an iOS app requests and uses location. The Android app was flagged under an Insecure Network Configuration, if it allowed unencrypted (cleartext) traffic or had a misconfigured Network security policy, and iOS was flagged if it had an App Transport Security (ATS) exception. Upon checking the establishment of WebView JavaScript on both the platforms, we were able to derive Web Attack Surface Metric followed by some browsable activities on Android or custom URL on iOS.

To enhance the privacy of each application four different indicators focusing on data exposure and tracking mechanism were evaluated. Tracker and SDK density were measured by aggregating the absolute counts of embedded trackers i.e Ad SDKs and Analytics SDKs. PII collections score was calculated as a count of different identifiable information

types such as email or device IDs. Lastly, we calculated a Sensitive Permission Count (Psens) which is an aggregate score that represents the app’s cumulative access to high risk hardware.

$$P_{sens} = \sum (Camera, Mic, Location, Contacts, SMS)$$

We formulated a Privacy Risk Index(PRI) using Principal Component Analysis (PCA) to do the cross platform comparison. PCA is used to combine the privacy indicators into a single metric. It captures maximum variance and avoids subjective weighting. PRI is derived from 8 factors: Tracker Count, Ad SDK Count, Analytics SDK Count, PII Count, Sensitive Permission Count, Advertising ID Usage, Insecure Network Configuration and Web Attack Surface.

First Principal Component (PC1) was extracted by standardizing the dataset using Standard Scaler and then processing via PCA. PC1 explained 37.22% of the total variance for the Android dataset and 36.48% variance for the iOS dataset. Final PRI score was then normalized to a 0-100 scale.

$$PRI = 100 \times \frac{PC_1 - \min(PC_1)}{\max(PC_1) - \min(PC_1)}$$

E. User-Perception vs Actual Risk

We collected recent user reviews (5000 reviews per application) from Google Play Store for analyzed Android applications, which were then analyzed using a privacy and security keyword based filtering like camera access, location tracking, microphone usage, etc.

We used VADER sentiment model to assign sentiment scores in the range [-1, +1] to each review and then normalized them to a 0-100 scale, where a higher value shows positive user perception. A neutral score was assigned to apps for which no relevant review was found.

IV. RESULTS

A. Cross Platform Privacy and Security Indicators

Fig. 2 shows the clear disparities that emerge between Android and iOS applications with respect to tracking-related components and any third-party SDK integrations.

Android applications generally exhibit a higher tracking footprint with an average tracking count of 4.64 per application as compared to iOS applications that show a near absence of trackers with a mean value of 0.14. Also, a similar pattern is observed for analytics SDKs, which are more commonly embedded in Android applications averaging 1.74 per app but are actively absent within the analyzed iOS samples. Our findings are consistent with prior work which shows that a substantial amount of Android applications transmit data to third-party advertising and analytics [24].

Prior research also shows over-permissioning in Android applications, particularly for network and personal data access [1]. Our results extend this observation by showing that Android applications collect a higher number of personally identifiable information categories with a mean of 3 compared to 2.2 for iOS applications.

These observations indicate that Android applications generally rely more heavily on tracking and analytics

infrastructures, while iOS applications exhibit a restrained presence of such components at the application level.

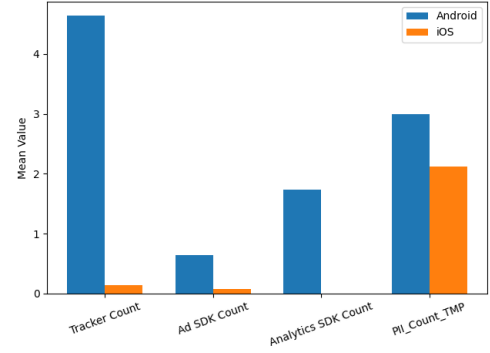


Fig. 2. Mean Privacy-Relevant Metrics

Camera permissions are present in 88.37% of iOS applications and 78.57% of Android applications. But there is a huge difference in microphone access, which is present in 79.07% of iOS applications and only 21.43% of Android applications.

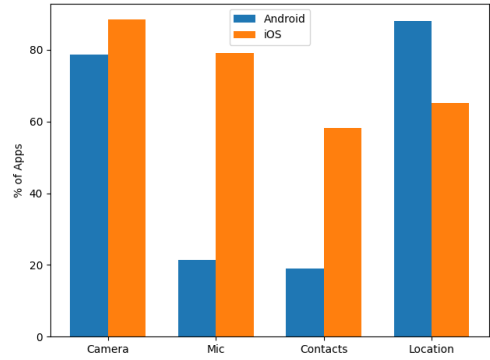


Fig. 3. Comparison of permissions used by applications

A similar trend can be observed in other permissions from Fig. 3. These results show that iOS applications request a wider range of direct user-facing permissions, whereas Android applications rely more on acquiring data through tracking and analytics mechanisms.

Behavioural exposures through embedded web components is widespread on both the platforms, though universal on iOS, where WebView JavaScript is enabled in all analyzed applications, compared to 80.95% on Android. This result aligns with prior work which shows that WebView is widely embedded in applications [25] and can introduce web-based attack surface.

Insecure network configuration is present in 47.62% of Android applications and 65.12% of iOS applications. Web-based attack surface affects 97.62% of Android applications and all analyzed iOS applications. iOS has stronger platform level security present but still many applications permit network behaviours that enhance the reachable attack surface. By this we can infer that regardless of the platform level security, how applications are implemented also becomes a deciding factor to security risks.

Android apps have an average PRI of 56.15, with a median of 49.42, whereas, for iOS, the average PRI is 63.31 and the median is 71.38 which is much higher than Android. From the boxplots in Fig. 4. we can observe that iOS apps cluster at

higher PRI values, whereas Android apps are more spread out, indicating more variation and a wider risk range.

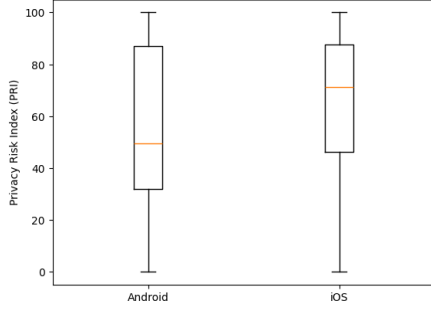


Fig. 4. PRI Distribution

Fig. 5 shows the factors on which the PRI depends. Access to sensitive permissions and the amount of personal information an app asks for, contribute to PRI much more than other factors.

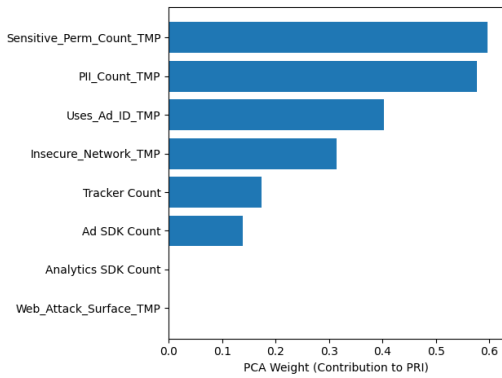


Fig. 5. Feature Contribution to Privacy Risk Index

While advertising identifiers and insecure network configuration presence also contribute to the PRI, factors like tracking counts and SDK related features contribute to a lesser extent.

B. Category-wise cross platform privacy risk

Apps were sorted into groups like, Travel and Food, Shopping and Ecommerce etc. Fig. 6 shows the category wise mean PRI values for both Android and iOS.

There is a lot of variation in risk scores depending on the category. Shopping & E-commerce and Social & Communications apps are riskier on iOS whereas health and finance apps have low PRI score for both platforms. Thus, we can see that some iOS app categories are more risky than others, despite Apple's closed ecosystem, whereas Android apps show more variation across categories.

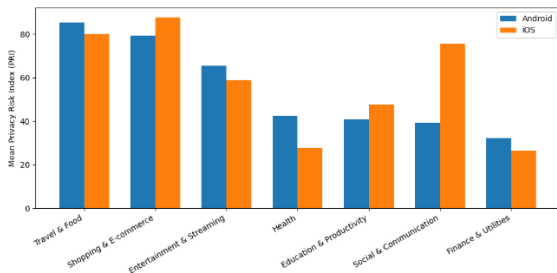


Fig. 6. Category-wise Privacy Risk: Android vs iOS

C. Pre-installed vs User-installed Android Applications

User-installed apps have 5.22 trackers on average, while pre-installed apps have just 0.5 trackers on average. Around 70% of pre-installed apps have insecure network security configuration present, which is 42% in user-installed apps.

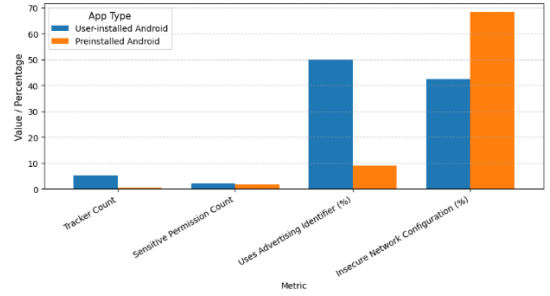


Fig. 7. Android apps: User-installed vs Pre-installed Comparison

Sensitive Permission Count is lesser in both user-installed and pre-installed applications, but pre-installed apps often request some permissions which are not dangerous but also not useful for what the app does. For example, Google calculator asks for full internet access, which is not a dangerous permission, but also of no use for the calculator to perform its intended functionality. Even if these permissions are normal, they still increase the privacy risk if the app doesn't need them.

D. iOS Apps Analysis

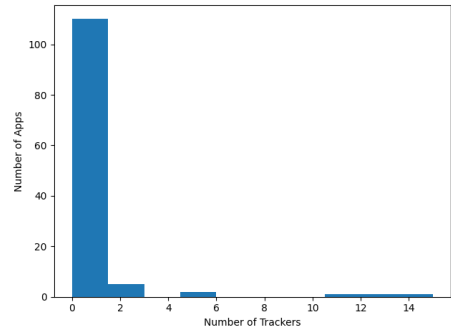


Fig. 8. Tracker Count Distribution in iOS Apps

Mostly iOS applications contain 0 or very few trackers, thus ensuring a restrictive tracking environment. Although, a few apps may still have a lot of third party trackers.

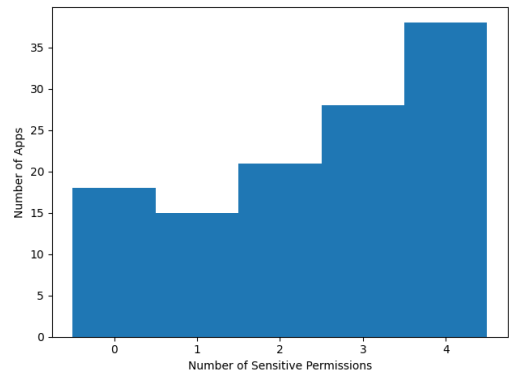


Fig. 9. Distribution of Sensitive Permissions in iOS Apps

Fig. 9 shows that iOS apps consistently access sensitive resources, which may potentially increase privacy risk.

E. Relation between static privacy risks and user sentiment

We observed strong negative correlations for Tracker count($\rho = -0.379$), Location Access($\rho = -0.346$) and WebView JavaScript Presence($\rho = -0.326$). Fig.10 shows that users are partially aware of camera usage and sensitive permission count, but overall these trust scores alone are not sufficient to indicate the technical privacy risks, reinforcing the need for objective security analysis.

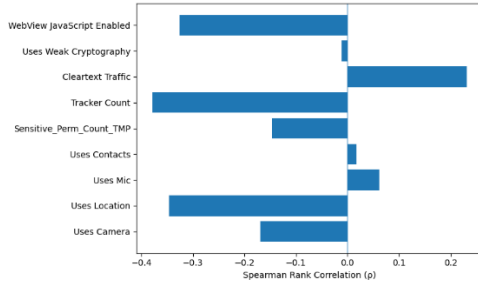


Fig. 10. Correlation between Static Privacy Risks and User Trust

V. CONCLUSION AND FUTURE WORK

We performed a cross platform evaluation of mobile application privacy and security using static analysis and Privacy Risk Index. The results show that privacy risk is not consistently determined by platform choice, rather it emerges from design trade-offs. Android apps reflected greater tracker density and collected many personally identifiable information. On the other hand, iOS apps have less number of trackers present but have a lot of user facing permissions like camera, microphone and contacts. This confirms that android is more inclined towards indirect data collection via third party ecosystems. Whereas, iOS depends more on sensor and permission access.

As future work, we plan to extend this comparison through dynamic analysis and observing runtime behaviours. Additionally, we plan to expand the dataset to include a much diverse set of applications from various categories, allowing us to further assess and evaluate both the platforms.

ACKNOWLEDGMENT

The authors used ChatGPT, an AI-based language model developed by OpenAI, for assistance with language refinement and formatting during the preparation of this manuscript. All technical content, analysis, and conclusions were reviewed and validated by the authors, who take full responsibility for the final content.

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